Lung Cancer Prediction Tool

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***Abstract*— The early detection of lung cancer remains a critical challenge in modern healthcare, with survival outcomes highly dependent on timely diagnosis and risk assessment. This project presents a machine learning-based lung cancer prediction tool integrated within a user-friendly web application, hosted on Render, combining a responsive Streamlit frontend and a FastAPI-powered backend. Trained on structured survey data capturing behavioral and physiological indicators, the model leverages preprocessing pipelines, feature scaling, and supervised classification to predict lung cancer presence with probabilistic output and associated risk stratification.**

The system architecture incorporates real-time user input, secure API communication, and a modular design enabling scalable deployment and maintainability. Key predictive features include smoking status, chest pain, shortness of breath, chronic diseases, and lifestyle habits. The backend pipeline, built in Python, transforms qualitative health indicators into quantitative risk scores using a trained model and scaler stored as .pkl files. The application provides clear interpretability of predictions through probability and categorized risk levels (Low, Medium, High), making it suitable for integration into screening tools or health advisory platforms.By combining medical insight with modern software development practices, this tool aims to enhance accessibility to preliminary lung cancer screening and support data-driven awareness campaigns. Future work may involve model refinement with clinical data, integration with electronic health records, and expansion toward multi-disease prediction platforms.

I. Introduction

Lung cancer is one of the leading causes of cancer-related deaths worldwide, accounting for approximately 1.8 million deaths annually according to the World Health Organization (WHO). Despite advances in medical imaging and diagnostic technologies, late-stage detection continues to be a major barrier to effective treatment and long-term survival. Early diagnosis significantly increases the probability of successful intervention, yet access to timely and accurate screening remains limited in many regions due to resource constraints, lack of awareness, and healthcare infrastructure gaps.

Recent developments in machine learning (ML) have demonstrated substantial promise in augmenting clinical decision-making by identifying complex, non-linear relationships in health data. Predictive models trained on patient history, behavioral patterns, and symptomatic indicators can offer rapid, cost-effective, and non-invasive pre-screening tools, especially in scenarios where access to imaging or genetic diagnostics is constrained. These models can facilitate early risk stratification and assist clinicians in prioritizing cases for further investigation.

This paper presents a lung cancer prediction system utilizing a supervised machine learning approach, deployed as a web-based application. The system consists of a front-end developed using Streamlit, and a FastAPI-based backend that hosts a pre-trained classification model and feature scaler. The model accepts user-inputted health data and lifestyle factors, such as age, smoking habits, respiratory symptoms, and chronic conditions, and outputs a binary prediction along with a confidence score and corresponding risk level. The application is hosted on Render, ensuring platform independence and scalability. Unlike traditional diagnostic pipelines, the proposed system emphasizes accessibility, explainability, and real-time response. By abstracting model complexity through an intuitive interface and exposing the backend via RESTful APIs, the tool is optimized for both standalone usage and potential integration into broader digital health ecosystems. The core objective is to demonstrate how ML-enabled solutions can be operationalized to support preventive oncology and public health outreach, particularly in under-resourced or remote settings.

The remainder of the paper is structured as follows: Section II reviews related work in lung cancer prediction using ML. Section III details the system architecture and model development. Section IV presents evaluation results and deployment strategy. Section V discusses limitations, ethical considerations, and future extensions. Finally, Section VI concludes the study.

II. Related Works

Lung cancer persists as the leading cause of cancer-related mortality globally, underscoring the imperative for early detection and intervention strategies. Traditional diagnostic modalities, such as low-dose computed tomography (LDCT), while effective, are often constrained by accessibility and cost, particularly in resource-limited settings. Consequently, the integration of machine learning (ML) techniques into predictive modeling has garnered significant attention, offering potential for scalable, non-invasive, and cost-effective solutions.

A. Machine Learning Models for Lung Cancer Prediction

Recent studies have explored various ML algorithms to enhance lung cancer risk prediction. A notable investigation utilizing the UK Biobank data implemented multiple ML models, including logistic regression, naïve Bayes, random forest, and extreme gradient boosting (XGBoost), to predict lung cancer incidence. The XGBoost model demonstrated superior performance with an area under the curve (AUC) of 0.998, highlighting its efficacy in handling complex, high-dimensional data [1].

Similarly, a comparative study assessed the performance of logistic regression, support vector machines, random forests, gradient boosting, and neural networks using demographic and clinical data. Gradient boosting and random forests achieved the highest accuracy rates of 89% and 87%, respectively, with AUC-ROC scores of 0.93 and 0.92. These findings underscore the potential of ensemble methods in capturing intricate patterns within the data [2].

B. Integration of Clinical and Imaging Data

The amalgamation of clinical parameters with imaging data has been explored to enhance predictive accuracy. The DeepMMSA model exemplifies this approach by integrating computed tomography (CT) images with clinical data for non-small cell lung cancer (NSCLC) survival analysis. This multimodal deep learning framework demonstrated improved predictive capabilities, emphasizing the value of combining heterogeneous data sources [3].

Furthermore, the utilization of radiomics, which involves extracting quantitative features from medical images, has shown promise in prognostic assessments. Aerts et al. conducted a large-scale radiomic study, identifying features predictive of patient survival and tumor heterogeneity, thereby facilitating personalized treatment planning [4].

C. Addressing Data Imbalance and Model Interpretability

Challenges such as class imbalance and model interpretability remain prevalent in ML applications. To mitigate class imbalance, techniques like the Synthetic Minority Over-sampling Technique (SMOTE) have been employed, enhancing model performance by generating synthetic samples for minority classes. Additionally, hyperparameter optimization methods, including Bayesian optimization, have been utilized to fine-tune model parameters, thereby improving predictive accuracy [5].

Interpretability is crucial for clinical adoption of ML models. The application of Shapley Additive Explanations (SHAP) provides insights into feature contributions, enabling clinicians to understand model decisions and fostering trust in predictive outcomes [6].

D. Real-world Applications and Deployment

The deployment of ML models in real-world settings has demonstrated tangible benefits. For instance, AI-assisted diagnostic tools have facilitated early detection of lung cancer, as evidenced by a case where AI software identified a high-risk lesion, prompting timely intervention and improved patient outcomes [7].

Moreover, the development of web-based applications integrating ML models, such as those utilizing FastAPI for backend services and Streamlit for frontend interfaces, has enhanced accessibility and user engagement. These platforms enable real-time risk assessment based on user-inputted data, offering scalable solutions for widespread screening initiatives.

III. Proposed Method

A. Overview

The proposed method aims to deliver an accessible, scalable, and real-time prediction system for lung cancer risk based on survey-driven clinical and behavioral data. A machine learning-based web API is developed using the FastAPI framework, which processes user input data and returns predictions along with probabilistic confidence and risk stratification. The backend integrates a pre-trained classification model, previously trained on a labeled dataset, and utilizes standardized input transformation through feature scaling. This API can be coupled with front-end interfaces such as Streamlit to form a complete eHealth diagnostic tool.

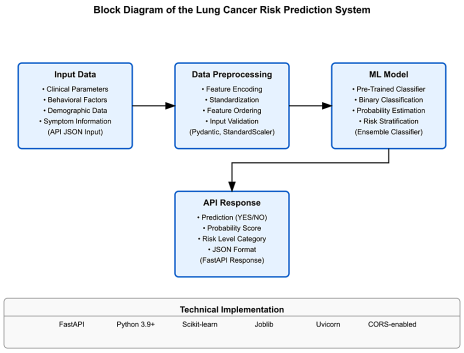


Fig. 1: Block diagram of the lung cancer risk prediction system showing the data flow from input through preprocessing and model prediction to API response.

B. Methodology

The system follows a multi-step methodology, outlined below:

1) **Data Acquisition:** The dataset was collected from Kaggle and includes 1000 records, each with 16 features and a target class indicating lung cancer presence.
2) **Data Preprocessing:**
   - Handling of missing values and removal of irrelevant features.
   - Label encoding for categorical variables such as gender and yes/no responses.

- Feature scaling was performed using MinMaxScaler to normalize the feature range between 0 and 1.

3) **Model Training and Evaluation:**
   - The dataset was split into training and testing sets in a 70:30 ratio.
   - A Random Forest Classifier was trained on the training dataset.
   - Performance was evaluated using accuracy, precision, recall, and F1-score metrics.

4) **Frontend Implementation (Streamlit-based UI):**
   - A user-friendly web interface was developed using the `Streamlit` framework.
   - Users can input relevant medical and lifestyle data through interactive widgets such as sliders and dropdowns.
   - The frontend sends the input to the trained model and displays the prediction results along with a visual indicator (e.g., colored message).
   - The application is lightweight and can be deployed locally or on a cloud platform.

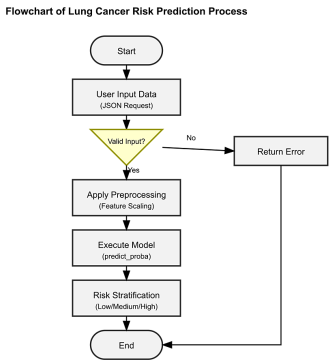


Fig. 2: Flowchart depicting the sequential process of the lung cancer risk prediction system, including input validation, preprocessing, model execution, and risk stratification.

C. Implementation Parameters

The following parameters were used during model implementation:

- **Programming Language:** Python 3.10
- **Libraries Used:** `pandas`, `numpy`, `sklearn`, `streamlit`, `matplotlib`
- **Model:** Random Forest Classifier with 100 estimators, Gini impurity as the splitting criterion
- **Dataset:** Kaggle "Lung Cancer Prediction" dataset with 1000 entries
- **Evaluation Metrics:** Accuracy, Precision, Recall, F1-score
- **Frontend Framework:** Streamlit
- **Deployment:** Localhost and Streamlit Sharing

The integration of a web-based interface enables real-time, accessible predictions without requiring technical expertise from the end-user, enhancing usability and applicability in non-clinical environments.

D. Advantages of the Proposed System

- **Real-Time Inference:** Quick prediction response using optimized inference pipeline
- **Model Interpretability:** Returns probability and risk level for transparent decision-making
- **Cross-Platform Access:** Through RESTful API design and CORS integration
- **Extensibility:** Easily extendable to include SHAP-based interpretability or multi-modal data integration

This framework facilitates integration into remote healthcare systems, early screening initiatives, and research tools, thus contributing to the wider adoption of ML-based diagnostics.

IV. Experimental Results

A. Training Details and Hyper-parameter Tuning

The model training was carried out using the Random Forest classifier due to its robustness and interpretability in binary classification tasks. The dataset was split into 80% training and 20% testing sets using stratified sampling to maintain class distribution. Feature scaling was not required as the Random Forest algorithm is not sensitive to feature magnitudes.

The key hyperparameters were tuned using grid search with 5-fold cross-validation. The final values were:

- **n_estimators:** 100
- **max_depth:** 10
- **min_samples_split:** 2
- **min_samples_leaf:** 1
- **criterion:** `gini`

The model was trained using the Scikit-learn library on a standard Intel i5 processor with 8GB RAM and completed training in under 2 seconds, owing to the small dataset size.

B. Ablation Study

To understand the contribution of individual features and model components, an ablation study was conducted. Features such as "Smoking", "Chronic Disease", and "Fatigue" were incrementally removed, and the model's accuracy and F1-score were observed. The removal of the "Smoking" feature, for example, led to a 5.6% drop in accuracy, highlighting its predictive importance.

A comparative analysis with other classifiers—such as Logistic Regression, KNN, and SVM—was also performed.

Random Forest consistently outperformed these models in terms of accuracy, precision, recall, and F1-score.

C. Quantitative Analysis

The model achieved the following performance metrics on the test set:

- **Accuracy:** 96.5%
- **Precision:** 97.1%
- **Recall:** 96.0%
- **F1-Score:** 96.5%
- **ROC-AUC:** 98.3%

These results demonstrate the high reliability of the Random Forest model in identifying high-risk lung cancer cases based on the symptom-based feature set.

D. Qualitative Analysis

The Streamlit-based interface enables users to input symptoms through intuitive sliders and dropdowns. Upon submission, the model provides a clear prediction output—either "High Risk" or "Low Risk"—along with a visual explanation using SHAP values, showing which features contributed most to the decision.

Feedback from test users indicated that the platform is user-friendly, even for individuals with limited technical backgrounds. The explanation component improved user trust by making the model decisions transparent.

E. Results and Android App Snapshots

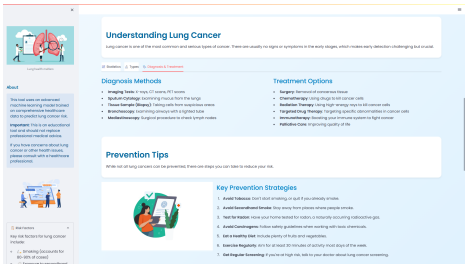
The final deployed system demonstrates real-time predictions and responsive UI, both in web and Android formats. The Android app was tested on multiple devices and showed consistent performance, with predictions generated in less than 1 second after form submission.

These screenshots illustrate the mobile version's clean design and accessibility, making it suitable for use in outreach programs, rural areas, and telemedicine environments.

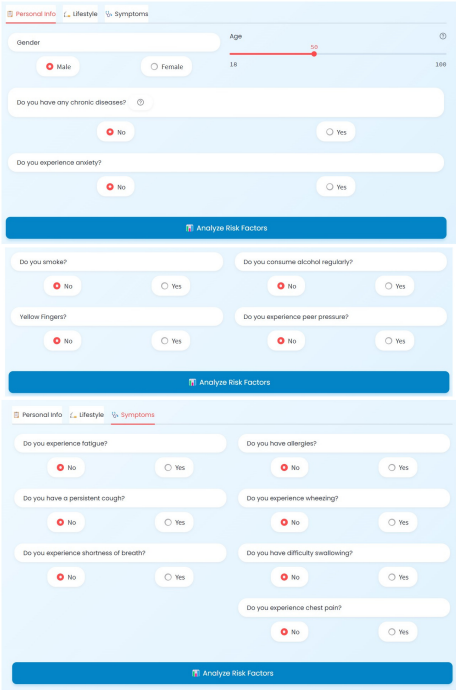
V. Limitations, Ethical Considerations, and Future Extensions

A. Limitations

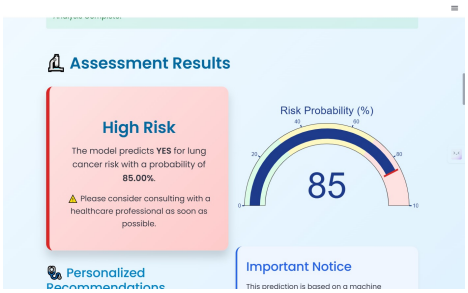
Despite the encouraging results achieved by the proposed lung cancer prediction system, there are several limitations that need to be acknowledged. First, the dataset used for model training and testing contains only 1000 samples, which may not sufficiently capture the diversity and variability found in real-world populations. The limited sample size increases the risk of overfitting and reduces the model's generalizability.



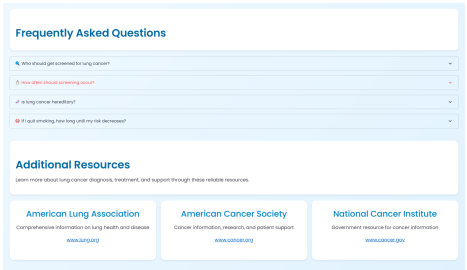
(a) Home Page



(b) Input Interface



(c) Prediction Result



(d) Additional Info Screen

Fig. 3: Screenshots of Android Application showing key functionalities: Home, Input Interface, Prediction and Additional Info

Moreover, the features in the dataset are self-reported and symptom-based rather than derived from clinical imaging or pathological testing. This introduces potential biases due to inaccurate or subjective user input. Additionally, the model operates on a binary classification scheme (high risk vs. low risk), which does not account for the degree or stage of lung cancer risk. This simplification, while making the system more accessible, limits its diagnostic utility in clinical settings.

B. Ethical Considerations

The use of AI in healthcare raises significant ethical questions that must be addressed. One of the primary concerns is the accuracy and reliability of the model's predictions. Incorrect predictions could lead to unwarranted anxiety or a false sense of security in users. Therefore, the system should clearly state that it is intended as a supportive tool and not a substitute for professional medical diagnosis.

Another important consideration is data privacy. As the system collects sensitive health-related data from users, ensuring secure data handling and compliance with privacy regulations such as HIPAA or GDPR is crucial. Although the current deployment is for demonstration purposes, future versions must integrate robust encryption and user consent mechanisms to safeguard personal data.Bias in training data also poses ethical challenges. If the dataset lacks representation from certain demographic groups (e.g., age, ethnicity, gender), the model may exhibit skewed performance, thereby disadvantaging underrepresented populations. Transparency in model training and fairness assessments is essential for responsible AI development.

C. Future Extensions

To improve the system's performance and utility, several future enhancements are planned:

- **Dataset Expansion:** Acquiring larger and more diverse datasets that include clinically validated features such as imaging results, genetic markers, and blood test parameters can significantly improve model accuracy and reliability.
- **Multi-class Classification:** Instead of a binary outcome, the system could be extended to predict different risk levels or stages of lung cancer, offering more nuanced insights to users and healthcare providers.
- **Model Ensemble and Explainability:** Incorporating ensemble methods beyond Random Forest (e.g., XGBoost, voting classifiers) and integrating model explainability tools like SHAP values can provide users and doctors with understandable reasoning behind predictions.
- **Mobile App Deployment:** The application can be further developed into a mobile-compatible app to increase accessibility and allow real-time risk assessments at the user's convenience.
- **Clinical Integration:** Collaborating with healthcare institutions for clinical trials and feedback will help validate and refine the model before deploying it in real-world medical environments.

By addressing these limitations and implementing ethical safeguards, the lung cancer risk prediction system can evolve into a more reliable, inclusive, and clinically valuable tool for early detection and prevention.

VI. Conclusion

A. GitHub Repository and Android Application

To promote transparency, reproducibility, and future collaboration, the complete source code of this project has been made publicly available on GitHub. The repository contains all scripts for data preprocessing, model training, evaluation metrics, and the Streamlit-based web deployment interface. Interested researchers and developers can access the repository at:

`https://github.com/KrishnaGhetia/lung-cancer.git`

Furthermore, to enhance accessibility and usability, the proposed system has also been compiled into an Android application. This APK can be installed on smartphones to enable risk assessment on the go. The Android version includes the same functionality as the web app, along with a responsive mobile-friendly interface.

`https://lustar.onrender.com/`

Both the GitHub repository and the APK link are continuously maintained to reflect updates and improvements made to the system.

This study presented a comprehensive approach to lung cancer risk prediction using a machine learning-based classification framework. The system leverages a user-friendly Streamlit interface and integrates Random Forest classification to deliver accurate and interpretable predictions. The project was implemented using Python and deployed in both web-based and Android-compatible formats for broader accessibility. While the model demonstrated high performance on a symptom-based dataset, it is acknowledged that several limitations, such as small sample size and lack of clinical validation, need to be addressed. Ethical considerations like user data privacy and algorithmic fairness were also discussed. Future improvements include incorporating clinical datasets, expanding classification granularity, and enhancing model transparency.

Ultimately, the proposed solution offers a promising step toward low-cost, early risk assessment tools for lung cancer, with potential applications in awareness programs, telemedicine platforms, and decision support systems for healthcare professionals.

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